David Battistelli

Email: david.battistelli@hotmail.it | Portfolio

EDUCATION

Nov. 2024 to Oct. 2026 — **Tuscia University** — **Viterbo, Italy**
MSc in Mechanical Engineering — *Final grade: 110/110 with honours*

- Subjects include Advanced Fluid Machinery and Energy Systems, Sensors and Data Acquisition system, Unconventional Technologies for Product Development, Nuclear Fusion
- Served as an elected student representative on the Engineering Faculty Council at Tuscia University.
- Master's thesis in collaboration with EPFL's Swiss Plasma Center on the development of a transferable framework that integrates the MEQ Suite with the TRUST fusion reactor and can be scaled to next-generation plasma devices.

Sep. 2021 to Nov. 2024 — **Tuscia University** — **Viterbo, Italy**
BSc in Industrial Engineering — *Final grade: 101/110*

- Contributed to R&D and chassis development for the university's Formula SAE team; the project received Tesla sponsorship for lithium-ion battery integration.
- Selected among the top 5% of candidates for a Porsche AG engineering project and factory visit in Stuttgart, based on academic performance and project submissions.
- Bachelor's thesis: "Analysis and Optimization of the Position of Stabilising Plates in the TRUST Reactor Experiment," using MATLAB and MAXFEA for simulation and design analysis.

WORK EXPERIENCE

Aug. 2025 to Nov. 2025 — **Swiss Plasma Center – EPFL** — **Lausanne, Switzerland**
Software Engineer & Project Manager Intern

- Worked on the integration of TRUST tokamak numerical models with MEQ (EPFL's proprietary suite for plasma equilibrium analysis), creating a framework compatible with and adaptable to a wide range of tokamaks.
- Contributed to the design of optimal configurations for magnetic coils and diagnostics to improve plasma profile reconstruction in TRUST.

Aug. 2023 to Oct. 2023 — **MG Sustainable Engineering AB** — **Uppsala, Sweden**
Project Management Intern

- Worked on the PV+ Solarus project by developing simulation code in Engineering Equations Solver and MATLAB to analyse the heat exchange in PVT panels; drafted a final report recommending 3+ material improvements and future design iterations
- Collaborated with Gävle University to model and simulate the electrical performance of PVT systems; leveraged energy laboratory equipment for detailed experimental analysis
- Assisted in designing electrical schematics for PVT panels using AutoCAD, producing 2D models to support system integration.

TECHNICAL SKILLS

- **Modelling & Analysis:** Numerical modelling and simulation, thermal analysis, data analysis.
- **Software & Tools:** Python (Intermediate), MATLAB (Intermediate), Engineering Equation Solver (Advanced), AutoCAD, Linux, Ansys (Basic), SolidWorks (Basic), MAXFEA (Basic), MS Office (Advanced).

ADDITIONAL INFORMATION

- Accredited as lifeguard by National pool lifeguard qualification in 2018.
- Selected in "Mentors4U" mentoring program among top 25% of applicants.
- Passionate about technology, geopolitics and sailing.
- Languages: Italian (Native), English (Fluent), Spanish and French (Intermediate)